

Course 12 — AI for Kids: Smart Projects

Ages 6–11 · 8 Lessons · 3 Portfolio Projects

Your child discovers how AI actually works, not by reading about it, but by building with it. Using Google's Teachable Machine and ML4Kids, they train machine learning models that recognize images and sounds, then connect them to Scratch to build interactive AI-powered projects any parent will be amazed by.

Why It Matters

AI already shapes the apps and devices kids use every day. This course demystifies it early: students see, hands-on, that AI is trained by people using real examples — not magic — and that they can build with it themselves. The core idea taught throughout: **Examples → Training → Model → Prediction → Action.**

Lesson Breakdown

1. What is AI and how do machines actually learn
2. Training your first image recognition model
3. Connecting an AI model to Scratch with ML4Kids
4. Building an AI rock-paper-scissors game
5. Voice commands: training a sound recognizer
6. Creating a voice-activated story in Scratch
7. Facial expression recognition and virtual pets
8. Showcase: presenting your AI project to the world

Portfolio Projects

1. **AI Rock-Paper-Scissors Game** — a real, complete Scratch game with score-keeping and full win/lose/tie logic.
2. **Voice-Activated Scratch Story** — a real branching interactive story with multiple scenes and a real choice point.
3. **Emotion-Recognition Virtual Pet** — a real virtual pet with moods, a Happiness meter, and animated reactions.

Important: each ships as a real, complete, playable `project.sb3` built entirely from standard Scratch blocks, using a simple keypress/click as a stand-in for the real AI signal (since a trained model's custom blocks are generated live, per-student, by Teachable Machine/ML4Kids and can't be pre-packaged). Each project's `AI_Training_Guide.md` and `Scratch_Integration_Guide.md` show exactly how to train the real model and wire it in.